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LIGHTING CONTROL APPARATUS WITH LINE SIDE CURRENT METERING

Abstract

A lighting control apparatus connectable to line conductors of an AC power source and a lamp driver includes a power metering circuit, a processor, and a communication circuit. The metering circuit is configured for coupling to the line conductors of the AC power source and the lamp driver. The processor is operably coupled to the power metering circuit and configured for coupling to the lamp driver. The metering circuit, which includes a current transformer arranged to detect AC current passing through the line conductor of the lamp driver, is operable to output a sensing signal corresponding to the AC current detected by the current transformer. The processor provides lighting control signals to the lamp driver, acquires the sensing signal from the power metering circuit, and processes the sensing signal into a format for communication. The communication circuit is operable to communicate the formatted sensing signal to a remote computing apparatus.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of and priority upon U.S. Provisional Patent Application No. 63/557,718, which was filed on Feb. 26, 2024, and is incorporated herein by this reference as if fully set forth herein.

TECHNICAL FIELD

[0002] The present disclosure relates generally to lighting control apparatus that receive utility power, such as from outdoor lighting fixtures, and, more particularly, to a lighting control apparatus with line side current metering functionality.

BACKGROUND

[0003] Lighting controllers for outdoor lighting fixtures are known. Such devices connect to a socket on a lighting fixture, such as a streetlight luminaire, and operate using batteries, solar cells, or alternating current (AC) power supplied to the lighting fixture by a utility. Some controllers are standalone, such as photocell sensors, and others are networked, such as those that support the Digital Addressable Lighting Interface (DALI) standards.

[0004] Some lighting controllers include metering functionality that measures AC line current using a very low resistance shunt resistor. However, such a measurement implementation typically includes a digital sampling circuit connected to the AC line directly, thereby necessitating additional circuitry and cost to isolate the AC signal from the digital sampling circuit.

SUMMARY

[0005] According to exemplary embodiments of the present disclosure, a lighting control apparatus includes a power metering circuit, a processor, and a communication circuit, and is connectable to at least the line conductors of an alternating current (AC) power source and a lamp driver of a lighting fixture (e.g., a driver of at least one light emitting diode (LED) module of an outdoor lighting fixture). The power metering circuit includes a current transformer arranged to detect AC current passing through the line conductor of the lamp driver and is operable to output a sensing signal corresponding to the AC current detected by the current transformer. The processor is operable to provide lighting control signals to the lamp driver, acquire the sensing signal from the power metering circuit, and process the sensing signal into a format for communication. The communication circuit is operable to communicate the formatted sensing signal to a remote computing apparatus, such as another lighting control apparatus, an edge processor, a centralized computing system, or a cloud server.

[0006] According to an alternative exemplary embodiment, the lighting control apparatus is further connectable to a neutral conductor of the AC power source and a neutral conductor of the lamp driver. Additionally or alternatively, the communication circuit includes a cellular modem (e.g., an LTE or 5G modem). According to further alternative exemplary embodiments, the power metering circuit has a metering accuracy of about $\pm 0.5\%$, and the lighting control apparatus is usable with an AC power source having a root mean square (rms) voltage in a range of approximately 120V to approximately 480V.

[0007] According to another alternative exemplary embodiment, the lighting control apparatus includes at least a pair of housing members and at least one printed circuit board (PCB). In this embodiment, the PCB(s) includes the power metering circuit, the processor, and the communication circuit, and is positioned upon at least a portion of a first side of one of the housing members (e.g., a base housing member). The other housing member (e.g., a cover housing

member) is positioned over at least the PCB(s) and optionally over a portion of the base housing member.

[0008] According to an alternative or additional exemplary embodiment, a second side of the base housing member includes at least one sidewall section along a periphery thereof and two or more spaced apart tab members extending from the sidewall section(s) toward a center of the second side of the base housing member. The tab members may cooperate with gaps and notches of a mounting bracket to secure the lighting control apparatus to a structure, such as an outdoor light fixture.

[0009] According to a further alternative exemplary embodiment, the base housing member defines an aperture therethrough and the lighting control apparatus further includes a conduit extending from the second side of the base housing member. In such an embodiment, the conduit defines a passageway in fluid communication with the aperture in the base housing member. Additionally or alternatively, the PCB is configured such that the processor and some or all components of the communication circuit, and even some components of the power metering circuit, are attached to one side of the PCB and at least the current transformer is attached to the other side of the PCB. In such an embodiment, the PCB is positioned upon at least a portion of the first side of the base housing member (e.g., upon the sidewall section(s) or a ledge of the sidewall section) such that at least part of the current transformer extends into the passageway of the conduit, or the current transformer is positioned on the PCB such that the current transformer extends into the passageway of the conduit when the PCB is positioned upon at least a portion of the first side of the base housing member.

[0010] According to other alternative exemplary embodiments, AC line, AC neutral, and/or lighting control signal conductors (e.g., wires) pass through the passageway of the conduit between their applicable end points. For example, the line conductor and/or the neutral conductor of the AC power source pass through the passageway of the conduit to the power metering circuit. Additionally or alternatively, the line conductor and/or the neutral conductor of the lamp driver pass through the passageway of the conduit from the power metering circuit. Additionally or alternatively, conductors carrying lighting control signals, which may be compliant with one or more Digital Addressable Lighting Interface (DALI) protocols, pass through the passageway of the conduit to the lamp driver.

[0011] According to other exemplary embodiments of the present disclosure, a lighting control apparatus includes a power metering circuit, a processor, and a communication circuit, and is connectable to at least the line conductors of an AC power source of a streetlight and a driver for an LED module of the streetlight. The power metering circuit includes a current transformer arranged to detect AC current passing through the line conductor of the LED driver and is operable to output a sensing signal corresponding to the AC current detected by the current transformer. The processor is operable to provide lighting control signals to the LED driver, acquire the sensing signal from the power metering circuit, and process the sensing signal into a format for wireless communication. The communication circuit is operable to wirelessly communicate the formatted sensing signal to a remote computing apparatus, such as another lighting control apparatus, an edge processor, a centralized computing system, or a cloud server.

[0012] According to another exemplary embodiment, the lighting control apparatus is further connectable to a neutral conductor of the AC power source and a neutral conductor of the LED driver. Additionally or alternatively, the communication circuit includes a cellular modem. According to further alternative embodiments, the power metering circuit has a metering accuracy of about $\pm 0.5\%$, and the lighting control apparatus is usable with an AC power source having a voltage in a range of approximately 120V AC (rms) to approximately 480V AC (rms).

[0013] According to another alternative exemplary embodiment of the present disclosure, the lighting control apparatus further includes at least two housing members and at least one PCB. In this embodiment, the PCB(s) includes the power metering circuit, the processor, and the communication circuit, and is positioned upon at least a portion of a first side of one of the housing

members (e.g., a base housing member). Another one of the housing members (e.g., a cover housing member) is positioned over at least the PCB(s) and optionally over a portion of the base housing member.

[0014] Alternatively or additionally, a second side of the base housing member includes at least one sidewall section along a periphery thereof and two or more spaced apart tab members extending from the sidewall section(s) toward a center of the second side of the base housing member. The tab members may cooperate with gaps and notches of a mounting bracket to secure the lighting control apparatus to a structure, such as an outdoor light fixture.

[0015] According to a further alternative exemplary embodiment, the base housing member defines an aperture therethrough and the lighting control apparatus further includes a conduit extending from the second side of the base housing member, wherein the conduit defines a passageway in fluid communication with the aperture of the base housing member. Additionally or alternatively, the PCB is configured such that the processor and some or all components of the communication circuit, and even some components of the power metering circuit, are attached to one side of the PCB and at least the current transformer is attached to the other side of the PCB. In such a case, the PCB is positioned upon at least a portion of the first side of the base housing member (e.g., upon the sidewall section(s) or a ledge of the sidewall section) such that at least part of the current transformer extends into the passageway of the conduit, or the current transformer is positioned on the PCB such that the current transformer extends into the passageway of the conduit when the PCB is positioned upon at least a portion of the first side of the base housing member.

[0016] According to other alternative embodiments, line, neutral, and/or lighting control signal conductors (e.g., wires) pass through the passageway of the conduit between their applicable end points. For example, the line conductor and/or the neutral conductor of the AC power source pass through the passageway of the conduit to the power metering circuit. Additionally or alternatively, the line conductor and/or the neutral conductor of the LED driver pass through the passageway of the conduit from the power metering circuit. Additionally or alternatively, conductors carrying the lighting control signals, which may be compliant with one or more DALI protocols, pass through the passageway of the conduit to the LED driver.

[0017] According to a further alternative exemplary embodiment of the present disclosure, a lighting control apparatus connectable to at least a line conductor of an AC power source and a line conductor of a lamp driver of a lighting fixture includes a lighting control device and a mounting bracket. In this embodiment, the lighting control device includes a power metering circuit, a processor, a communication circuit, a first housing member (e.g., a base housing member), a second housing member (e.g., a cover housing member), and at least one PCB. The power metering circuit includes a current transformer arranged to detect AC current passing through the line conductor of the lamp driver and is operable to output a sensing signal corresponding to the AC current detected by the current transformer. The processor is operable to provide lighting control signals to the lamp driver, acquire the sensing signal from the power metering circuit, and process the sensing signal into a format for communication. The communication circuit is operable to communicate the formatted sensing signal to a remote computing apparatus, such as another lighting control apparatus, an edge processor, a centralized computing system, or a cloud server. The PCB(s) includes the power metering circuit, the processor, and the communication circuit, and is positioned upon at least a portion of a first side of the base housing member. The cover housing member is positioned over at least the PCB(s) and optionally over a portion of the base housing member.

[0018] The mounting bracket is securable to a structure, such as a luminaire of a streetlight, a lamp or other portion of a lighting fixture, or a lamp post or pole. Where the structure is a streetlight luminaire, the luminaire may include the lamp driver to which the lighting control device provides lighting control signals and a lamp (e.g., LED module) driven by the lamp driver. The mounting bracket defines two or more gaps and includes two or more notches along a periphery thereof,

wherein each notch is adjacent to a respective gap. The width of each gap permits passage of a tab member of the base housing member. Each notch rests upon a respective tab member after the tab member has passed through the respective gap adjacent to the notch and the lighting control device has been rotated so as to position the notch over the tab member.

[0019] Although the present disclosure illustrates and describes several exemplary embodiments for a structure-mountable electronic device assembly, the disclosure is, nevertheless, not intended to be limited to the details shown because various modifications and structural changes may be made to the disclosed embodiments without departing from the scope and range of equivalents of the appended claims. Additionally, well-known elements of the disclosed embodiments will not be described in detail or will be omitted so as not to obscure the relevant details of such embodiments.

[0020] Detailed exemplary embodiments of an electronic device assembly are disclosed herein; however, the disclosed embodiments are merely exemplary, and the assembly may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one of ordinary skill in the art to variously employ the claimed invention in exemplary embodiments. Further, the terms and phrases used herein are not intended to be limiting but rather are intended to provide an understandable description of the present disclosure.

[0021] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The terms “a” or “an,” as used herein, mean one or more. The term “plurality,” as used herein, means two or more. The term “another,” as used herein, means at least a second or more. The terms “comprises,” “includes,” “contains,” “has,” and their respective formatives, as used in the present disclosure and the appended claims, are intended to be open-ended or non-exhaustive (i.e., open language) and should be interpreted as if each was followed by the words “but is not limited to.” The terms “coupled” and “connected,” as used herein, mean connected directly or indirectly; permanently, semi-permanently, or temporarily; mechanically, integrally, electrically, logically, or operably; or in any other manner. The term “providing” is defined herein in its broadest sense (e.g., bringing/coming into physical existence, making available, and/or supplying to someone or something, in whole or in multiple parts at once or over a period of time).

[0022] As used in the present disclosure, unless otherwise specified, relationships indicated by terms such as “up,” “down,” “left,” “right,” “inside,” “outside,” “interior,” “exterior,” “front,” “back,” “head,” “tail,” “base,” “cover,” “top,” “bottom,” and so on, are azimuth or positional relationships based on the drawings or to identify elements or objects, and are only intended to facilitate the descriptions of the disclosed embodiments of the present disclosure, but not to indicate or imply that the elements or objects must have a specific azimuth, or be constructed or operated in the specific azimuth. Furthermore, terms such as “first,” “second,” “third,” and so on are only used for identification purposes and should not be construed as indicating or implying any relative importance or order.

[0023] As used in the present disclosure and the appended claims, the term “longitudinal” should be understood to mean in a direction corresponding to an elongated direction of an object.

[0024] The terms “about,” “substantially,” “generally,” or “approximately” apply to all numeric values, whether explicitly indicated or not. When used expressly or impliedly in the present disclosure and the appended claims, such terms refer to a range of values, quantities, features, and/or functionality that one of ordinary skill in the art would consider equivalent to the recited values, quantities, features, and/or functionality (e.g., would provide an equivalent result). In many instances these terms may include numbers that are rounded to the nearest significant figure. Those skilled in the art will readily understand the specific meanings of the above-mentioned terms in the embodiments of the present disclosure according to the specific circumstances.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views and which together with the detailed description below are incorporated in and form part of the specification, serve to further illustrate various embodiments and explain various principles and advantages all in accordance with the present disclosure. Some of the elements and features disclosed herein may be viewable in one or more of the figures, but not necessarily in all the figures. The figures of the drawings are not drawn to scale.

[0026] FIG. 1 is an exploded view of an electronic device assembly, in accordance with exemplary embodiments of the present disclosure.

[0027] FIG. 2 is a bottom view of an exemplary mounting bracket for use in the electronic device assembly of FIG. 1.

[0028] FIG. 3 is a bottom view of the electronic device assembly of FIG. 1 in assembled form.

[0029] FIG. 4 is an elevational view of an exemplary electronic device assembly as mounted in an exemplary manner to a vertical (e.g., decorative) lighting fixture with a partial cut-away view of a base of the lighting fixture to show an optional threaded external surface of the electronic device assembly, in accordance with some alternative embodiments of the present disclosure.

[0030] FIG. 5 is a perspective view of an electronic device assembly as mounted in an exemplary manner to a support structure, such as a cobra head luminaire, in accordance with some exemplary embodiments of the present disclosure.

[0031] FIG. 6 illustrates an exploded perspective view of an exemplary printed circuit board (PCB) module for an electronic device of the electronic device assembly of FIG. 1 and a top perspective view of the PCB retaining apparatus of the PCB module, in accordance with exemplary embodiments of the present disclosure.

[0032] FIG. 7 illustrates, in PCB layout and block diagram form, a top view of an exemplary PCB for the PCB module illustrated in FIG. 6.

[0033] FIG. 8 is a top view of the exemplary PCB retaining apparatus of FIG. 6.

[0034] FIG. 9 is a bottom view of the exemplary PCB retaining apparatus of FIG. 6.

[0035] FIG. 10 is a top view of the exemplary PCB retaining apparatus of FIG. 6 as attached to a base housing member of an exemplary electronic device forming part of the electronic device assembly of FIG. 1.

[0036] FIG. 11 is a perspective view of a partially assembled electronic device with a partially or fully translucent cover housing member and the top section of the cover housing member excluded for clarity purposes, in accordance with other exemplary embodiments of the present disclosure.

[0037] FIG. 12 is a cross-sectional view of the electronic device of FIG. 11 with the top section of the cover housing member included.

[0038] FIG. 13 is an enlarged cutaway view of a portion of an interior surface of the electronic device of FIG. 11 illustrating exemplary facets included on the interior surface.

[0039] FIG. 14 is an electrical block diagram of a lighting control device including a light sensing system and a power metering circuit, in accordance with further exemplary embodiments of the present disclosure.

[0040] FIG. 15 is an electrical block diagram of an exemplary power metering circuit of the lighting control device of FIG. 14.

[0041] FIG. 16 is a bottom perspective view of the electronic device assembly of FIG. 1 in assembled form.

[0042] FIG. 17 is a bottom plan view of the assembled electronic device assembly of FIG. 16.

DETAILED DESCRIPTION

[0043] FIG. 1 is an exploded view of an electronic device assembly, in accordance with exemplary

embodiments of the present disclosure. The electronic device assembly **100** includes an electronic device **190** and a mounting bracket **104**. The electronic device **190** includes a first or base housing member **101**, a printed circuit board (PCB) **102** with a plurality of electrical and optionally other components arranged and configured to enable the electronic device **190** to perform its intended function, and a second or cover housing member **103**. The base housing member **101** may be positioned to form the bottom housing member of the electronic device **190** and the cover housing member **103** may be positioned to form the top housing member of the electronic device **190**, or vice versa. The mounting bracket **104** is described in more detail below with respect to FIG. 2.

[0044] According to some embodiments, the electronic device **190** further includes a PCB retaining apparatus **105**, which together with the PCB **102** and its associated components **149** form a PCB module **192**. As illustrated in more detail in FIGS. 6, 8, and 9, the PCB retaining apparatus **105**, when included, includes a support member **601** having a first side **612** and an opposing second side **614**, a plurality of flexible finger members **602-604** (three numbered for illustrative purposes but two or more could be used) positioned proximate a periphery **605** of the support member **601**, and at least one wall member **606-611, 902-920** (several numbered for illustrative purposes but one or more may be used depending on several factors, such as PCB size, PCB layout, PCB component types and quantities, configuration of the electronic device, and so forth). Each finger member **602-604** extends away from the support member **601** in a direction generally opposite the first side **612** of the support member **601** and has a restoring force toward a center of the second side **614** of the support member **601** (or, more generally, toward a center of the support member **601**). The restoring forces of the finger members **602-604** cause inside surfaces of the finger members **602-604** to engage peripheral areas **640** of the PCB **102** (e.g., at least the edges) to retain the PCB **102** in place after it has been positioned between the finger members **602-604**. Each wall member **606-611, 902-920** extends away from the second side **614** of the support member **601** and is sized and shaped to restrict deflection of at least one area of the PCB **102** responsive to one or more externally applied forces after the PCB **102** is positioned between the finger members **602-604**. For example, the wall members **606-611, 902-920** are configured to at least restrict deflections of one or more areas of the PCB **102** containing electrical or mechanical components **149, 622, 702, 704, 1444** of the PCB **102**. Advantageously, the retaining apparatus **105** provides a robust structure to help minimize shock and potential damage to the PCB **102** and its components in the event the electronic device assembly **100** is exposed to one or more undesirable external forces, such as impacts from being dropped or having objects or projectiles contacting it.

[0045] As illustrated in FIGS. 6 and 9, the wall members **606-611** may be individually or collectively configured into various shapes as necessary to restrict deflections in selected areas of the PCB **102**. For example, wall member **606** may generally form an “L shape” with a long member **902** and a short member **904**. Additionally or alternatively, wall members **607-609** may form a multi-wall member **906** configured generally in a “C shape.” Further, another multi-wall member **908** may include a base wall member **918** and four radial wall members **911-914**. Still further, another wall member **920** may be configured generally in a “U shape.” The quantity of anti-deflection wall members and their configurations will depend on the configuration and component locations of the particular PCB **102**.

[0046] The cover housing member **103** is illustrated in FIG. 1 as being generally dome-shaped (e.g., a compound dome, a geodesic dome, an oval dome, or a generally semi-spherical shape) but may be shaped otherwise (e.g., paraboloid, parallelepiped, conical, etc.) with an open bottom **181**, one or more sidewalls (which may have an optional upper sidewall portion **182** and an optional lower sidewall portion **183**), and a top **185** (which may be at least partially translucent). Interior surfaces of the sidewall(s) and the top **185** form an interior chamber that accommodates at least the PCB module **192**. In some embodiments, the upper sidewall portion **182** is also translucent.

[0047] According to some exemplary embodiments, the exterior surfaces **186** of the top **185** and the upper sidewall portion **182** are generally smooth while the external surface of the lower

sidewall portion **183** is irregular, such as faceted, grooved, abrasive, rough, wavy, or curved. When so configured, the generally irregular exterior surface **187** of the lower sidewall portion **183** provides an enhanced gripping surface for twisting or rotating the electronic device **190** during mounting or dismounting of the electronic device **190** to or from the mounting bracket **104** at the time of installation or removal, as appropriate.

[0048] The base housing member **101** has a first side **110** (e.g., a top side) and an opposing second side **111** (e.g., a bottom side or underside). According to some exemplary embodiments, the top side **110** of the base housing member **101** includes at least one sidewall section **121** along an outer periphery thereof, wherein an outside surface **122** of the at least one sidewall section **121** defines a channel **123** configured to receive a seal, such as a gasket, an O-ring **161**, or a sealant, at least partially therein. The at least one sidewall section **121** may be a single continuous sidewall, a series of connected sidewall sections, or a set of spaced sidewall sections with gaps in between.

[0049] According to some embodiments, the channel **123**, when included, is configured to beneficially receive and retain the seal therein. The seal helps mitigate the likelihood that moisture will enter the electronic device **190** or reach the PCB **102** or its components located inside the chamber of the cover housing member **103**, such as during inclement weather (such as rain, sleet, or snow, for example). When used, the seal preferably fits snugly against the interior surface of the sidewall of the cover housing member **103**, such as at or about where the upper sidewall portion **182** meets the lower sidewall portion **183**.

[0050] According to additional or alternative embodiments, one or more retaining clips **171**, **172** (two shown for illustrative purposes in FIG. 1) are positioned on or along an inside surface **173** of the at least one sidewall section **121** to retain at least a portion of the PCB **102**. In such embodiments, one or more peripheral areas of the PCB **102** are positioned into the clips **171**, **172** (e.g., pressed or slid into the clips **171**, **172**) such that the PCB **102** is secured by the clips **171**, **172** to, for example, a ledge along the at least one sidewall section **121** of the first side **110** of the base housing member **101**. Positioning peripheral areas of the PCB **102** on a ledge allows the PCB **102** to be double-sided, having components on both sides. Alternatively, where the PCB **102** has components on only one side (e.g., the top side), the retaining clips **171**, **172** and the PCB **102** may be positioned further down the inside surface(s) **173** of the at least one sidewall section **121** and closer to an aperture **113** defined by and through the base housing member **101**.

[0051] According to some embodiments as illustrated in FIG. 1, the electronic device **190** further includes a conduit **112** extending from the second side **111** of the base housing member **101**. When included, the conduit **112** defines a passageway **133** in fluid communication with the aperture **113** defined by and passing through the base housing member **101**. The conduit passageway **133** is sized and shaped to allow wires and/or conductors **114**, such as the line conductor of an alternating current (AC) power source, the AC line conductor of a lamp driver, one or more ground conductors, direct current (DC) power conductors, and control signal wires, just to name a few, to be fed through it to and/or from the PCB **102**. The shape of the conduit **112** is also such that it passes through a central opening **127** in the mounting bracket **104**. To provide water sealing, the passageway **133** may be filled with epoxy after the wires are threaded through the passageway **133** or the wires **114** may be secured in a plug that has its own O-ring or other seal to snugly fit into the passageway **133** to provide a water-tight seal.

[0052] In other exemplary embodiments, the bottom side **111** of the base housing member **101** is configured to be twist-lockable with the mounting bracket **104**, as described in more detail below with respect to FIGS. 2, 3, 16, and 17. In these embodiments, the mounting bracket **104** is first attached to a support structure, such as, for example, a utility pole, a streetlight luminaire or other lighting fixture, a wall of a building, or any other structure having electrical power accessible from it, and then the electronic device **190** is secured in a twist-lock manner to the mounting bracket **104**.

[0053] According to some exemplary embodiments, the cover housing member **103** is at least

partially translucent to allow ambient light impinging on the exterior surface of the cover housing member **103** to pass through the translucent portion of the cover housing member **103** to either a light sensor **622** (e.g., a photosensor) on the PCB **102** or an entrance or input **126** to an optional light guide **620** of or coupled to the PCB retaining apparatus **105** (described in more detail below with respect to FIG. 6), which light guide **620** directs the light (or at least some of it) to the light sensor **622** on the PCB **102**. In one exemplary embodiment in which the electronic device **190** is a lighting control apparatus (or a lighting control device thereof) or otherwise includes lighting control functionality, the light sensor **622** provides an output signal corresponding to the amount of incident light detected, which output signal is used by a processor of the electronic device **190** to generate lighting control signals for a driver of a light source (e.g., one or more LED panels of a lighting fixture). For example, when the output signal of the light sensor **622** indicates the luminosity of the incident light is at or above a predetermined threshold, the processor will not produce a control signal that causes the driver to turn on the light source or, if the light source is already on, the processor will produce a control signal that causes the driver to turn off the light source. On the other hand, when the output signal of the light sensor **622** indicates the luminosity of the incident light is below the predetermined threshold, the processor will not produce a control signal that causes the driver to turn off the light source or, if the light source is already off, will produce a control signal that causes the driver to turn on the light source.

[0054] FIG. 2 illustrates a bottom view of an exemplary embodiment of the mounting bracket **104**. The exemplary mounting bracket **104** defines a plurality of gaps **141-145** and includes a plurality of notches **151-155** along an outer periphery portion of the bracket **104**. As may be better understood with reference to FIGS. 2, 16, and 17, each notch **151-155** is adjacent to a respective gap **141-145**, and a width **146** of each gap **141-145** permits passage of a respective tab member **131a-131e** on the bottom side **111** of the base housing member **101**. Each notch **151-155** of the mounting bracket **104** rests upon a respective tab member **131a-131e** of the base housing member **101** after the tab member **131a-131e** has passed through the respective gap **141-145** adjacent to the notch **151-155** and the base housing member **101** or the entire electronic device **190** has been rotated so as to position each tab member **131a-131e** over its associated notch **151-155** of the mounting bracket **104**.

[0055] The electronic device assembly **100** may be securely mounted to a support structure, such as a utility pole, a streetlight pole, a streetlight luminaire, or other structure from which electrical power is accessible. To do so, the mounting bracket **104** is first mounted to the support structure. Such mounting may be performed using one or more fasteners or by other means. For example, referring to FIG. 5, where the electronic device assembly **100** is intended for mounting to a streetlight luminaire **502** that has a National Electrical Manufacturers Association (NEMA) socket in compliance with the American National Standards Institute (ANSI) C136 series of standards, the screws that would otherwise be used to secure the NEMA socket to the luminaire may be instead used to secure the mounting bracket **104** to the luminaire at the same locations as the NEMA socket would be normally mounted. Alternatively, the electronic device assembly **100** may be mounted to a streetlight luminaire **502** at the location where an external connector in compliance with another roadway lighting connection standard, such the Zhaga Book 18 standard, would otherwise be connected.

[0056] After the mounting bracket **104** has been mounted to the support structure, the base housing member **101** of the electronic device **190** is secured to the mounting bracket **104**. The base housing member **101** may be attached to the mounting bracket **104** by, for example, aligning the tab members **131a-131e** of the base housing member **101** with the gaps **141-145** of the mounting bracket **104**, moving the base housing member **101** toward the mounting bracket **104** such that tab members **131a-131e** pass through the gaps **141-145** in the mounting bracket **104**, and rotating the base housing member **101** or the entire electronic device **190** relative to the mounting bracket **104** until each notch **151-155** is positioned over and/or rests upon a respective tab member **131a-131e**

of the base housing member **101**. According to some embodiments, the base housing member **101** or the entire electronic device **190** may be rotated relative to the mounting bracket **104** until the leading side edges of the tab members **131a-131e** of the base housing member **101** engage the closed ends or edges of the notches **151-155** of the mounting bracket **104**. According to some embodiments, the widths of the gaps **141-145** of the mounting bracket **104** and the widths of the tab members **131a-131e** of the base housing member **101** may be mutually varied (as opposed to all being the same) so as to form a keyed system for mounting the base housing member **101** to the mounting bracket **104**, particularly where orientation of the electronic device **190** as mounted to the structure is critical to operation of the electronic device **190**. In general, the electronic device assembly **100** is configured to facilitate a simple and repeatable process of mounting the assembly **100** to a structure, especially under circumstances in which installers may be wearing gloves and operating in a bucket truck for safety purposes.

[0057] According to some embodiments of the present disclosure, the mounting bracket **104** may be a low profile disc or flat circular structure with a central opening **127** configured to receive the conduit **112** of the electronic device **190** through which wires or conductors **114** are routed. Additionally, the mounting bracket **104** may include a grooved or recessed area **147** proximate the central opening **127** to receive a seal (e.g., an O-ring **148**, gasket, or adhesive) positioned between the underside **111** of the base housing member **101** and the mounting bracket **104**. The outer periphery of the mounting bracket **104** may also include arched sections **135-139** adjacent to the notches **151-155**, where each arched section **135-139** defines or includes a slotted aperture **128** and an outer protrusion **129**. During mounting of the electronic device **190** to the mounting bracket **104** or removal of the electronic device **190** from the mounting bracket **104**, the slotted apertures **128** allow the protrusions **129** of the mounting bracket **104**, when included, to deflect slightly inward during rotation of the electronic device **190** or its base housing member **101**. Referring also to FIGS. **16** and **17**, the protrusions **129** then return to a non-deflected or normal position to lock or mate into or with complementarily shaped recesses **184** on the inside surface(s) of the sidewall section(s) **120** on the bottom side **111** of the base housing member **101**.

[0058] The exemplary electronic device **190** and the exemplary electronic device assembly **100** are shown in assembled forms from various viewpoints in FIGS. **3**, **16**, and **17**. For example, FIG. **3** is a bottom plan view of the assembled electronic device assembly **100**, FIG. **16** is a bottom perspective view of the assembled electronic device **190**, and FIG. **17** is a bottom plan view of the assembled electronic device **190**. According to some embodiments, as illustrated in FIG. **3**, the bottom side **111** of base housing member **101** is mounted to the mounting bracket **104** so as to maintain a low profile along a Z-axis with reference to the 3D axis system **130** shown in FIG. **1**.

[0059] In some embodiments as illustrated in FIGS. **16** and **17**, the bottom side **111** of the base housing member **101** includes a plurality of spaced apart tab members **131a-131e** extending from the inside surface of the sidewall section(s) **120** toward the center of the bottom side **111** of the base housing member **101** or, more generally, toward the center of the base housing member **101**. Additionally or alternatively, the inside surface(s) of the sidewall sections(s) **120** also includes recesses **184** that are sized and shaped to receive protrusions **129** of the mounting bracket **104** when the electronic device **190** is rotatably secured to the mounting bracket **104**.

[0060] According to other embodiments as illustrated in FIG. **3**, the cover housing member **103** may include a plurality of optional spaced apart retaining clips **371-373** (three shown for illustration) along a bottom edge thereof. The retaining clips **371-373** include tabs or lips that extend toward the center of the cover housing member **103** and engage a bottom edge of the base housing member's sidewall **120** or bottom edges of one or more of the base housing member's sidewall sections (e.g., where the sidewall **120** is not continuous) when the cover housing member **103** is positioned over the PCB **102** and pushed onto the base housing member **101**. When included, the retaining clips **371-373** have restoring forces that cause the cover housing member **103** to lock to the base housing member **101**.

[0061] FIG. 4 illustrates an alternative exemplary electronic device assembly 401 as mounted in an exemplary manner to a vertical lighting fixture and includes a partial cut-away view of a base 404 of the lighting fixture to show an optional threaded external surface 410 of the electronic device assembly 401, in accordance with some alternative embodiments of the present disclosure. In this case, the electronic device 401 is substantially similar to the electronic device 190 described above, except that the conduit 112 is threaded (i.e., includes threads 410) on an exterior surface thereof. The support structure 404 to which the electronic device 401 is attached may have mating threads along an interior surface thereof at the electronic device attachment point. For these embodiments, the electronic device assembly 401 includes the electronic device and the threads 410 of the conduit instead of the electronic device 190 and the mounting bracket 104.

[0062] In some exemplary embodiments, the electronic device 190 includes a controller or other processor and provides Internet of Things (IoT) functionality. The IoT functionality can include power metering, lighting control, and asset management, as well as transmitting associated data via wireless communication to a remote computer system or server to facilitate reporting and alerting. In one exemplary embodiment, the electronic device 190 is a networked lighting controller in a smart lighting system.

[0063] FIG. 6 illustrates an exploded bottom perspective view of the PCB module 192 for the electronic device 190 of the electronic device assembly 100 of FIG. 1 and a top perspective view of the PCB retaining apparatus 105 of the PCB module 192, in accordance with exemplary embodiments of the present disclosure. According to such embodiments and referring also to FIGS. 7-10, the retaining apparatus 105 includes a support member 601, a plurality of flexible finger members 602-604 (three shown for illustration) positioned proximate a periphery 605 of the support member 601, and at least one wall member 606-611 (six identified by reference numerals for illustration). The support member 601 has a first side 612 (e.g., top side) and an opposing second side 614 (e.g., bottom side or underside). Each finger member 602-604 extends away from the support member 601 in a direction generally opposite the top side 612 of the support member 601 and has a restoring force toward a center of the bottom side 614 of the support member 601 or, more generally, toward a center of the support member 601. The one or more wall members 606-611 also extend away from the bottom side 614 of the support member 601, such as in the direction opposite the first side 612 of the support member 601. The wall members 606-611 are sized and shaped to restrict deflection of at least one area of the PCB 102 responsive to an externally applied force after the PCB 102 is positioned (e.g., inserted, push fit, or otherwise installed or assembled) between the finger members 602-604.

[0064] In one exemplary embodiment and referring also to FIG. 8, the support member 601 includes a substantially flat section 630 and an angled section 631 joined to the substantially flat section 630, where the angled section 631 extends to the periphery 605 of the support member 601. In some embodiments, the flat section 630 of the support member 601 has a generally circular shape as illustrated in FIG. 6. However, persons of ordinary skill in the art will readily recognize and appreciate that the flat section 630 may be any other desired shape depending on the geometric design of the PCB retaining apparatus 105, including oval, oblong, square, rectangular, pentagonal, hexagonal, octagonal, and the like.

[0065] According to some exemplary embodiments, the PCB retaining apparatus 105 (or at least the support member 601, the finger members 602-604 and the one or more wall members 606-611) is formed as a single molded part or component. Beneficially, a single molded component can provide a mass-producible and cost-effective design. When formed as a molded component, the PCB retaining apparatus 105 may be fabricated from a plastic that provides structural integrity while adding desired flexibility. Some exemplary materials from which to mold the PCB retaining apparatus 105 include acrylonitrile butadiene styrene (ABS), polycarbonate, a polycarbonate and ABS blend, polyphenylene oxide, polyphenylene ether, polybutylene terephthalate, polyamide, polymethyl methacrylate, styrene ethylene butadiene styrene, and acrylic.

[0066] In some exemplary embodiments, the support member **601** (e.g., the flat section **630** thereof) defines an aperture located at least partially above an expected location of a light sensor **622** secured to the PCB **102**. When included, the aperture is sized and shaped to accommodate and partially receive a light guide **620** therethrough to direct incident light toward the light sensor **622**. The light sensor **622** (shown in phantom in FIG. **6**) is surface mounted or otherwise secured to the top side **624** of the PCB **102**, for example. An entrance or input **126** of the light guide **620** may be located on the top side **612** of the support member **601** of the PCB retaining apparatus **105**. In alternative embodiments, the light guide **620** may be integrated with and formed as part of the PCB retaining apparatus **105**. In some embodiments, the light guide **620** extends from the underside **614** of the PCB retaining apparatus **105** toward the top side **624** of the PCB **102** such that the light guide's output is positioned just above (e.g., 2-25 mm above) the light sensor **622** on the PCB **102**.

[0067] In further exemplary embodiments, some or all the finger members **602-604** include tabs **615** on inward-facing sides **617** of the finger members **602-604**. When included, the tabs **615** provide support for the PCB **102** upon positioning of the PCB **102** between the finger members **602-604** or equivalently upon positioning of the finger members **602-604** about the PCB **102**. In one such embodiment, each finger member **602-604** includes a tab **615** on its inward-facing side **617** for enhanced connections to and support of the PCB **102**. In an alternative embodiment, at least three finger members **602-604** include tabs **615** on their inward-facing sides **617**.

[0068] In some exemplary embodiments, the PCB **102** further includes an outer periphery or outer peripheral areas **640** defining one or more recesses **638** (two shown for illustration in FIG. **6** and three shown for illustration in FIG. **7**) that are complementarily configured to allow one or more longer finger members **676** with inside-facing tabs **625** and restoring forces toward the center of the PCB retaining apparatus **105** to pass through the spaces created by the recesses **638** and engage an underside **626** of the PCB **102** with the tabs **625**. When properly assembled, each longer finger member **676** resides partially in a respective recess **638**, and each tab **625** contacts the bottom side **626** of PCB **102** for an enhanced connection and support.

[0069] In some exemplary embodiments, multiple wall members **607-611** may be used to restrict deflection of the PCB **102** near larger PCB components, such as larger integrated circuits **702**, **704**, **1444**, capacitors **642**, inductors, transformers, toroids, and so forth. In such cases, the walls **607-611** are positioned adjacent to and/or just above the component to minimize and restrict unwanted deflection of the area or areas of the PCB **102** at or to which the components **642**, **702**, **704**, **1444** are attached (e.g., soldered) in the event of an undesirable externally applied force or mechanical shock, for example.

[0070] According to other exemplary embodiments in which the electronic device **192** transmits or receives information wirelessly, such as a through a cellular modem **702** or another wireless communication circuit, or receives GNSS signals (e.g., GPS signals) to facilitate location determination, the electronic device **192** includes one or more antennas **632**, **634** for such purposes. According to the exemplary embodiments illustrated in FIGS. **6**, **8**, and **10**, the antennas **632**, **634** are positioned (e.g., disposed or deposited on, attached to, secured to, and the like) at least partially on the top side **612** of the support member **601**. Alternatively, the antennas **632**, **634** may be positioned wholly or partially on the bottom side **614** of the support member **601**. Further, depending on how grounding is arranged in the electronic device **192**, a portion of each antenna **632**, **634** may be disposed on one or more of the longer fingers **676** (or posts) to allow the antenna **632**, **634** to be electrically connected to a ground plane on the top side **624** or the bottom side **626** of the PCB **102**, for example. According to one exemplary embodiment, antenna **632** supports the cellular modem **702** illustrated in block form in FIG. **7** in connection with cellular communications, such as LTE or 5G communications, and antenna **634** supports reception of GPS signals for processing by the cellular modem **702** (where such modem **702** is GPS-enabled) or the electronic device's main processor **704**, as also illustrated in block form in FIG. **7**. As should be understood by those skilled in the art, other signaling protocols may be used and supported by the antennas **632**,

634 depending on the application or use case. Additionally, multiple antennas may be used for a single purpose (e.g., diversity antennas or multiple input multiple output (MIMO) antennas may be used for cellular communications).

[0071] FIGS. **11-13** illustrate various views of an electronic device **1100** in accordance with further alternative exemplary embodiments of the present disclosure. The electronic device **1100** includes several elements of the electronic device **100** of FIG. **1** and may include all the elements of such electronic device **100**, except with modifications to the cover housing member **103** as described in more detail below.

[0072] According to the alternative embodiments illustrated in FIGS. **11-13**, the electronic device **1100** includes a light sensing system **1102** (shown generally in block form in FIG. **11**) and an at least partially translucent housing member **1104** positioned over at least the light sensing system **1102** and, more preferably, over the entire PCB **1120** (e.g., PCB **102**) containing the light sensing system **1102**. The housing member **1104** is shown without its top in FIG. **11** merely to aid the relevant description of the electronic device **1100**. Similar to the cover housing member **103** illustrated in FIG. **1**, the housing member **1104** may be generally dome-shaped (e.g., a compound dome, a geodesic dome, an oval dome, or a generally semi-spherical shape) or shaped otherwise (e.g., paraboloid, parallelepiped, conical, etc.) with an open bottom, one or more at least partially translucent sidewalls, and an at least partially translucent top (not shown). Thus, the housing member **1104** defines a chamber within its interior to house the PCB **1120** and its components (or at least the components attached to the top side of the PCB **1120**), including the components of the light sensing system **1102**. In some embodiments, the sidewall(s) has a translucent upper sidewall portion **182** and an optional lower sidewall portion **183**, which may also be translucent. The lower sidewall portion **183** may have an irregular exterior surface to improve grip when mounting the electronic device **1100** to a mounting bracket **104**, for example.

[0073] The sidewall(s) of the electronic device **1100** have an interior surface **1106** and an exterior surface **1108**. In the embodiments illustrated in FIGS. **11-13**, the interior surface **1106** includes a plurality of facets **1110** configured to direct light impinging on the exterior surface **1108** of the translucent section of the sidewall(s) (e.g., the upper sidewall portion **182** of the housing member **1104**) toward the light sensing system **1102** or a portion of it, such as the light sensor **622** or the entrance or input **126** to a light guide **620** or light pipe (e.g., where the PCB **1120** is retained by a retaining apparatus **105** and the retaining apparatus **105** includes, uses, or accommodates the light guide **620**). Advantageously, the facets **1110** are designed and configured to create a prismatic or Fresnel lens effect that focuses/converges light rays **1112** passing through the translucent upper sidewall portion **182** of the housing member **1104** to a predetermined or desired target or location within the chamber of the housing member **1104**, such as toward the light sensing system **1102** or the input **126** to a light guide **620** having its output positioned just above (e.g., 2-25 mm above) the light sensor **622**, for example.

[0074] The housing member **1104** illustrated in FIGS. **11-13** may be the cover housing member **103** of the electronic device **100** shown in FIG. **1**, with an interior surface **1106** including facets **1110** as illustrated in FIGS. **11-13**. As shown in enlarged form in FIG. **13**, the facets **1110** of the housing member's interior surface **1106** are arranged in pairs in some embodiments and together with a respective translucent portion of the exterior surface **1108** forms a triangular prism that directs the impinging light toward its intended target (e.g., the light sensor **622** or the light guide input **126**). In alternative embodiments, the facets **1110** are integrated into the interior surface **1106** of the housing member **1104** but may alternatively be formed into a separate film or covering that is adhered to the interior surface **1106** of the housing member **1104**. In further embodiments, the facets **1110** are evenly distributed along the interior surface **1106** of the housing member **1104**, such as illustrated in FIGS. **11** and **12**. In other embodiments, each facet **1110** is angled relative to a reference axis, such as an axis substantially perpendicular to the PCB **1120** when the electronic device **1100** is fully assembled, so as to more accurately or effectively direct light impinging on the translucent

exterior surface **1108** to the location of the intended target (e.g., the light sensor **622** or the input **126** to the light pipe **620**). In some exemplary embodiments, such as illustrated in FIG. **13**, each facet **1110** is angled in the range of approximately three degrees (3°) to approximately five degrees (5°) relative to a respective reference axis and toward the center of the electronic device **1100** (e.g., relative to the Z-axis in the reference 3D axis system **130** shown in FIGS. **1** and **12**) for a more consistent and enhanced focus of impending light on the desired target. In further exemplary embodiments, the exterior surface **1108** of the housing member **1100**, or at least the exterior surface of the upper housing portion **182**, is smooth and devoid of any facets. Where the electronic device **1100** is intended for use in outdoor environments and, more specifically, is a lighting control device, a smooth exterior surface **1108** reduces the adherence of debris to the exterior surface **1108**, which could affect operation of the light sensing system **1102** by preventing light from entering one or more translucent areas of the housing member's exterior surface **1108**.

[0075] In some embodiments, the electronic device **1100** includes a processor **1116** (e.g., main processor **704**) coupled to the light sensing system **1102**. The processor **1116** is operable to, among other things, generate a lighting control signal responsive to receipt of a light sensing signal from the light sensing system **1102**. The lighting control signal may be communicated to and be usable by a lamp driver of a lamp, such as an LED driver of one or more LED modules in an outdoor lighting fixture (e.g., a streetlight). For example, the lighting control signal may instruct the lamp driver to turn on the lamp, dim the lamp, or turn off the lamp.

[0076] In other embodiments, the electronic device **1100** includes a second housing member, such as the base housing member **101** of the electronic device **190** described above with respect to FIGS. **1-10**. In such embodiments, the PCB **1120** is positioned upon at least a portion of the first side **110** of the base housing member **101**, such as a lip or ledge of the base housing member **101**, especially where the PCB **1120** has components mounted or attached to both of its sides.

[0077] In further exemplary embodiments, the electronic device **1100** includes the PCB retaining apparatus **105** (and optionally its light guide **620**), as well as various other components and elements of the electronic device **190** of FIG. **1**. Such components and elements are described in detail above with respect to FIGS. **1-10** and will not be repeated here for brevity.

[0078] In some exemplary embodiments in which the electronic device **1100** includes the PCB retaining apparatus **105**, the top side **612** of the support member **601** is configured and strategically located within the chamber defined by the housing member **1104** such that light impinging on the exterior surface **1108** in the upper sidewall portion **182** can be directed by the facets **1110** to the input **126** of the light pipe **620** and thereby to the light sensor **622**. Advantageously, such a structure and strategic placement permits the directed light rays **1112** to be substantially free from interference in reaching the light guide input **126**, thereby contributing to the efficiency of the light sensing system **1102**.

[0079] According to some other alternative embodiments as illustrated in FIG. **12**, the interior surface **1106** of the housing member **1104** may include several functional sections **1122-1128**. For example, one section **1122** may include the facets **1110**. Another section **1124** may be adjacent to the PCB **1120**. A third section **1126** may be adjacent to the channel **123** containing the seal (e.g., the O-ring **161**). The last section **1128** may be adjacent to the periphery of the base housing member **101**. In one embodiment, three interior surface sections **1124-1128** are configured to substantially complement the mechanical structures of the adjacent areas of the PCB **1120**, the seal channel **123** or associated sidewall of the base housing member **101**, and the periphery of the base housing member **101** to provide a snug and secure fit and an enhanced interface after the housing member **1104** is assembled over the PCB **1120** and at least part of the base housing member **101**.

[0080] FIG. **13** illustrates, in enlarged form, a section of the interior surface **1106** of the cover housing member **1104** of the electronic device **1100** to more clearly show exemplary positioning of the interior surface facets **1110** relative to the PCB **1120** and an edge or lip **1301** of the base housing member **101**. For example, the interior surface facets **1110** may rest on or against the edge

or lip **1301** of the base housing member **101** and adjacent to a periphery of the PCB **1120** after the housing member **1104** is positioned over the PCB **1120** and secured to the base housing member **101**.

[0081] FIG. **14** is an electrical block diagram for an exemplary lighting control device **1400**, such as a networked lighting controller, with line current metering capability, in accordance with some embodiments of the present disclosure. The lighting control device **1400** includes a metering circuit **1416**, a main processor (e.g., main processor **704**), a communication circuit **1430** with associated antenna **1480**, and a light sensor **622** (e.g., a photosensor). The lighting control device **1400** may optionally include a power regulator circuit **1404** that includes DC-to-DC voltage conversion when the DC power used to create the supply voltage **1448** for the metering circuit **1416**, the main processor **704**, the communication circuit **1430**, and any other circuitry that relies upon such voltage **1448** is unregulated or at too high of a voltage. The components of the lighting controller **1400** may include some or all the electrical components and circuits disposed on or secured to the PCB **102** of the electronic device **100** described above with respect to FIGS. **1-10** or disposed on or secured to the PCB **1120** of the electronic device **1100** described above with respect to FIGS. **11-13**. The metering circuit **1416** is described in more detail below with respect to FIG. **15**.

[0082] In some exemplary embodiments as illustrated in FIGS. **14** and **15**, a lighting control apparatus, such as or which may include lighting control device **1400**, is connectable to at least a line conductor of an alternating current (AC) power source (e.g., utility power supplied to a utility or streetlight pole) and at least a line conductor of a lamp driver, such as an LED driver of a streetlight luminaire. According to some embodiments, the lighting control apparatus **1400** includes a power metering circuit **1416**, a processor **704**, and a communication circuit **1430**. In other embodiments, the lighting control apparatus **1400** includes a light sensor **622** and associated circuitry to form an ambient light sensor (ALS.) The metering circuit **1416** is configured for coupling to at least the line conductor of the AC power source and at least the line conductor of the lamp driver. According to some embodiments such as the embodiment illustrated in FIG. **14**, the metering circuit **1416** is further configured for coupling to the neutral conductor of the AC power source and the neutral conductor of the lamp driver to pass AC power from the AC power source to the lamp driver. In some embodiments, the metering circuit **1416** is configured to meter AC power with a root mean square (rms) voltage in a range of approximately 120 VAC to approximately 480 VAC. Such range of AC power is generally used in connection with typical lower power use cases, such as in streetlights, at residential service points, and at light commercial service points.

[0083] The main processor **704** is a processor or controller with moderate processing power sufficient to acquire a sensing signal from the metering circuit **1416**, process the sensing signal into a format for communication by the communication circuit **1430**, and generate and provide lighting control signals to the lamp driver to which the main processor **704** is electrically connected based on, for example, an output of the light sensor **622** or a light sensing system **1102** that includes the light sensor **622**. In some embodiments, the communication circuit **1430** includes a cellular modem **702** (e.g., an LTE modem or a 5G modem) to provide wireless communication over a cellular network. Alternatively or additionally, the communication circuit **1430** may include other communication modems or transceivers to facilitate communication over wired or other wireless networks. The main processor **704** supplies the appropriately formatted sensing signal acquired from the metering circuit **1416** to the communication circuit **1430** to communicate the formatted sensing signal and its embedded metering data to a remote computing system, such as a monitoring and management software platform executing on a cloud server. The main processor **704** may also communicate other data and information via the communication circuit **1430**, including operating data and lamp status data, as well as receive through the communication circuit **1430** lighting control signals, firmware updates, and other instructions from the remote computing system.

[0084] According to some exemplary embodiments as illustrated in FIG. **15**, the power metering circuit **1416** includes a current transformer **1418** and a metering processor **1444**. The power

metering circuit is arranged to detect AC current passing through the line conductor of the lamp driver and output a sensing signal corresponding to the AC current detected by the current transformer **1418**. The metering processor **1444** processes and stores the output voltage of the current transformer **1418**, which is directly proportional to the AC current passing through the conductor around which the current transformer **1418** is positioned. Use of a current transformer **1418** instead of a shunt resistor to detect AC current eliminates the need for a digital sampling circuit to be connected to the AC line, thereby also eliminating the need for additional circuitry and cost to isolate the AC signal from the digital sampling circuit to ensure accurate current readings. Use of the current transformer **1418** enables the power metering circuit **1416** to have a metering accuracy of about $\pm 0.5\%$.

[0085] According to some further exemplary embodiments, the lighting control apparatus **1400** further includes a power regulator circuit **1404**, such as where the metering circuit **1416**, the main processor **704**, the communication circuit **1430**, and other components of the lighting control apparatus **1400** requiring a direct current (DC) supply voltage **1448** rely on an unregulated DC power source or a DC power source providing a voltage greater than necessary to supply the DC supply voltage **1448**. In such embodiments, the power regulator circuit **1404** may provide a DC-to-DC voltage conversion, such as a down conversion from the DC source voltage to the DC supply voltage **1448** used by the main processor **704**, the metering circuit **1416**, and the communication circuit **1430**, as well as perform voltage regulation. In one embodiment, the power regulator circuit **1404** receives DC power from a lamp driver (e.g., LED driver) of a streetlight. Where the lamp driver operates in accordance with one or more Digital Addressable Lighting Interface (DALI) standards, such as the DALI D4i standard, and supplies an auxiliary output voltage, the voltage supplied by the lamp driver to the power regulator circuit **1404** may be about 24V DC. The supply voltage **1448** output by the power regulator **1404** may be about 3V DC or such other voltage as required by the metering circuit **1416**, the main processor **704**, the communication circuit **1430**, and other electrical components of the lighting control apparatus **1400**.

[0086] According to some exemplary embodiments, the metering circuit **1416**, the main processor **704**, the communication circuit **1430**, and the light sensor **622** form part of a lighting control device **1400** in a multi-component lighting control apparatus. According to one exemplary embodiment, the components of the lighting control device **1400** are attached to one or both sides of a printed circuit board, such as the PCB **102** or the PCB **1120**. For example, as illustrated in FIGS. **6** and **7**, some components of the lighting control device (e.g., the main processor **704**, the communication circuit **1430** (e.g., cellular modem **704**), the light sensor **622**, and the metering processor **1444**) may be attached to one side of the PCB **102**, **1120** and the current transformer **1418** and other components may be attached to the other side of the PCB **102**, **1120**. Alternatively, the components of the lighting control device **1400** may be attached to multiple PCBs or on just one side of the PCB **102**, **1120**.

[0087] In embodiments where the current transformer **1418** is attached to the bottom side of the PCB **102**, **1120** as illustrated in FIGS. **6** and **12**, the current transformer **1418** is preferably positioned on the PCB **102**, **1120** such that at least part of the current transformer **1418** resides in the passageway **133** of the conduit **112** of the lighting control device (e.g., electronic device **190**). Depending on the configuration of the current transformer **1418** to handle the AC voltage and current ranges expected from the AC power source, the height of the current transformer **1418** may be such that the component cannot fit in the space between the bottom side of the PCB **102**, **1120** and a floor on the top side **110** of the base housing member **101**. Therefore, positioning the current transformer **1418** on the bottom side of the PCB **102**, **1120** such that at least part of the current transformer **1418** extends into the passageway **133** provides sufficient clearance for the current transformer **1418** while allowing the lighting control device to maintain a low profile and the current transformer **1418** to be positioned on the side of the lighting control device at which the line conductor current from the AC power source enters the lighting control device (e.g., through

the passageway **133**).

[0088] In further exemplary embodiments in which the lighting control apparatus or its lighting control device is configured similar to the configuration of the electronic device **190**, **1100** illustrated in FIG. **1** or FIG. **11**, the line and/or neutral conductors (e.g., insulated or bare wires) connecting the AC power source to the metering circuit **1416** and connecting the metering circuit **1416** to the lamp driver pass through the passageway **133** of the conduit **112**. For example, the line and neutral conductors of the AC power source pass through the passageway **133** of the conduit **112** to the power metering circuit **1416** and the line and neutral conductors of the lamp driver pass through the passageway **133** of the conduit **112** from the power metering circuit **1416**. In alternative embodiments, lighting control signals from the main processor **704** (e.g., lighting controller) pass through the passageway **133** of the conduit **112** to the lamp driver. Where the lamp driver is compliant with one or more of the DALI protocols (e.g., DALI-2 and/or D4i), the lighting control signals are compliant with the applicable DALI protocol or protocols. Where the lamp driver is not DALI-compliant, the lighting control signals may be other conventional lighting control signals, such as pulse wave modulation signals or analog voltage control signals (e.g., 0-10V dimming).

[0089] In some alternative embodiments, the metering processor **1444** may be optionally coupled to an infrared (IR) LED **1466**. In such cases, the metering processor **1444** controls the IR LED to report the metering data in real time for on-location detection by an IR sensor (e.g., a handheld sensor used by installation or maintenance personnel).

[0090] According to further alternative embodiments, a lighting control apparatus includes the lighting control device **1400** described above with respect to FIGS. **14** and **15**, as packaged in the form of the electronic device **190** of FIG. **1** or the electronic device **1100** of FIG. **11**. Therefore, according to such embodiments, the lighting control apparatus includes the lighting control device **1400** together with the mounting bracket **104**. As so arranged, the lighting control apparatus may be mounted to a structure, such as a streetlight luminaire that includes a lamp driver (e.g., LED driver) and a lamp (e.g., LED module(s)) driven by the lamp driver.

[0091] The claims appended hereto are meant to cover all modifications and changes within the scope and spirit of the present disclosure.

Claims

1. A lighting control apparatus connectable to at least a line conductor of an alternating current (AC) power source and a line conductor of a lamp driver of a lighting fixture, the lighting control apparatus comprising: a power metering circuit that includes a current transformer arranged to detect AC current passing through the line conductor of the lamp driver, the power metering circuit being operable to output a sensing signal corresponding to the AC current detected by the current transformer; a processor operable to provide lighting control signals to the lamp driver, acquire the sensing signal from the power metering circuit, and process the sensing signal into a format for communication to produce a formatted sensing signal; and a communication circuit operable to communicate the formatted sensing signal to a remote computing apparatus.
2. The lighting control apparatus of claim 1, wherein the lamp driver is a driver of at least one LED module.
3. The lighting control apparatus of claim 1, wherein the lighting control apparatus is further connectable to a neutral conductor of the AC power source and a neutral conductor of the lamp driver.
4. The lighting control apparatus of claim 1, wherein the communication circuit includes a cellular modem.
5. The lighting control apparatus of claim 1, wherein the AC power source has a root mean square (rms) voltage in a range of approximately 120V to approximately 480V.

6. The lighting control apparatus of claim 1, wherein the power metering circuit has a metering accuracy of about $\pm 0.5\%$.

7. The lighting control apparatus of claim 1, further comprising: a first housing member having a first side and an opposing second side; at least one printed circuit board positioned upon at least a portion of the first side of the first housing member, the at least one printed circuit board including the power metering circuit, the processor, and the communication circuit; and a second housing member positioned over at least the at least one printed circuit board.

8. The lighting control apparatus of claim 7, wherein the at least one printed circuit board is a single printed circuit board, wherein the processor and components of the communication circuit are attached to a first side of the printed circuit board, and wherein the current transformer is attached to a second side of the printed circuit board.

9. The lighting control apparatus of claim 8, wherein the first housing member defines an aperture therethrough, the lighting control apparatus further comprising: a conduit extending from the second side of the first housing member, the conduit defining a passageway in fluid communication with the aperture of the first housing member; wherein the printed circuit board is positioned upon at least a portion of the first side of the first housing member such that at least part of the current transformer extends into the passageway of the conduit.

10. The lighting control apparatus of claim 9, wherein the line conductor of the AC power source passes through the passageway of the conduit to the power metering circuit.

11. The lighting control apparatus of claim 9, wherein the line conductor of the lamp driver passes through the passageway of the conduit from the power metering circuit.

12. The lighting control apparatus of claim 1, wherein the lighting control signals are signals compliant with one or more Digital Addressable Lighting Interface (DALI) protocols.

13. A lighting control apparatus connectable to at least a line conductor of an alternating current (AC) power source of a streetlight and a line conductor of a driver for a light emitting diode (LED) module of the streetlight, the lighting control apparatus comprising: a power metering circuit that includes a current transformer arranged to detect AC current passing through the line conductor of the driver for the LED module, the power metering circuit being operable to output a sensing signal corresponding to the AC current detected by the current transformer; a processor operable to provide lighting control signals to the driver for the LED module, acquire the sensing signal from the power metering circuit, and process the sensing signal into a format for wireless communication to produce a formatted sensing signal; and a communication circuit operable to wirelessly communicate the formatted sensing signal to a remote computing apparatus.

14. The lighting control apparatus of claim 13, wherein the power metering circuit is configured for coupling to at least the line conductor and a neutral conductor of the AC power source and at least the line conductor and a neutral conductor of the driver for the LED module.

15. The lighting control apparatus of claim 13, further comprising: a first housing member having a first side and an opposing second side; at least one printed circuit board positioned upon at least a portion of the first side of the first housing member, the at least one printed circuit board including the power metering circuit, the processor, and the communication circuit; and a second housing member positioned over at least the at least one printed circuit board.

16. The lighting control apparatus of claim 15, wherein the at least one printed circuit board is a single printed circuit board, wherein the processor and components of the communication circuit are attached to a first side of the printed circuit board, and wherein the current transformer is attached to a second side of the printed circuit board.

17. The lighting control apparatus of claim 16, wherein the first housing member defines an aperture therethrough, the lighting control apparatus further comprising: a conduit extending from the second side of the first housing member, the conduit defining a passageway in fluid communication with the aperture of the first housing member; wherein the printed circuit board is positioned upon at least a portion of the first side of the first housing member such that at least part

of the current transformer extends into the passageway of the conduit.

18. The lighting control apparatus of claim 17, wherein the line conductor of the AC power source passes through the passageway of the conduit to the power metering circuit and wherein the line conductor of the driver for the LED module passes through the passageway of the conduit from the power metering circuit.

19. A lighting control apparatus connectable to at least a line conductor of an alternating current (AC) power source and a line conductor of a lamp driver of a lighting fixture, the lighting control apparatus comprising: a lighting control device including: a power metering circuit that includes a current transformer arranged to detect AC current passing through the line conductor of the lamp driver, the power metering circuit being operable to output a sensing signal corresponding to the AC current detected by the current transformer; a processor operable to provide lighting control signals to the lamp driver, acquire the sensing signal from the power metering circuit, and process the sensing signal into a format for communication to produce a formatted sensing signal; a communication circuit operable to communicate the formatted sensing signal to a remote computing apparatus; a first housing member having a first side and an opposing second side, wherein the second side of the first housing member includes at least one sidewall section along a periphery thereof and a plurality of spaced apart tab members extending from the at least one sidewall section toward a center of the first housing member; at least one printed circuit board positioned upon at least a portion of the first side of the first housing member, the at least one printed circuit board including the power metering circuit, the processor, and the communication circuit; and a second housing member positioned over at least the at least one printed circuit board; and a mounting bracket securable to a structure, the mounting bracket defining a plurality of gaps and including a plurality of notches along a periphery thereof, wherein each notch is adjacent to a respective gap, wherein a width of each gap permits passage of a tab member of the first housing member of the lighting control device, and wherein each notch rests upon a respective tab member after the tab member has passed through the respective gap adjacent to the notch and the lighting control device has been rotated so as to position the notch over the tab member.

20. The lighting control apparatus of claim 19, wherein the structure is a streetlight luminaire that includes the lamp driver and a lamp driven by the lamp driver.
